THE E_1 SPECTRAL SEQUENCE FOR A SIMPLICIAL TOPOLOGICAL SPACE WITH E_1^{P,Q} = H^Q(X_P, F_P)

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: functoriality in F (statement line 632) rides on the cited Homology first-quadrant spectral sequence lemma, as recorded for the module-twin card L46178; this lemma is the (2d) ingredient of card L46177 (augmentation-pushforward-higher-direct-image); the identification A^{p,q} = Mor_PAb(Z_{X_p}, I^q) is at source lines 637-641.

Editorial note. The label lemma-simplicial-sheaf-cohomology names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Setup: a simplicial topological space and an abelian sheaf. Statement lines 624-630. X a simplicial topological space, F an abelian sheaf on X_{Zar}; the spectral sequence is claimed with E_1^{p,q} = H^q(X_p, F_p), functorial in F.
- The Z-resolution on the simplicial site. Same chapter, \label{lemma-simplicial-resolution-Z} and Cohomology on Sites, Lemma \label{sites-cohomology-lemma-injective-abelian-sheaf-injective-presheaf}. Proof lines 634-641: A^{p,q} = Gamma(X_p, I_p^q) = Mor_PAb(Z_{X_p}, I^q); the rows of the double complex are exact in positive degrees and evaluate to Gamma(X_{Zar}, I^q) in degree 0.
- Exactness of restriction and injectivity of the restricted sheaves. Same chapter, \label{lemma-restriction-to-components} (card L46149) and \label{lemma-restriction-injective-to-component}. Proof lines 641-648: restriction is exact, so F_p -> I_p^bullet is a resolution; the I_p^q are injective abelian sheaves on X_p.
- The homological-algebra machinery. Homology, Lemmas \label{homology-lemma-first-quadrant-ss} and \label{homology-lemma-double-complex-gives-resolution}. Proof lines 648-651: the first-quadrant spectral sequence lemma and the double-complex resolution lemma conclude.

Source reference. Original lemma 0133: sources/stacks-chow/tags.txt, source lines 1421–1421. Prerequisite pointer invoked at proof line(s) 665; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0132: sources/stacks-chow/tags.txt, source lines 1420–1420. Prerequisite pointer invoked at proof line(s) 664; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09VM: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 352–352. Prerequisite pointer invoked at proof line(s) 642; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09W1: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 492–492. Prerequisite pointer invoked at proof line(s) 661; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09W0: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 456–456. Prerequisite pointer invoked at proof line(s) 654; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09W5: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 590–590. Prerequisite pointer invoked at proof line(s) 648; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03F6: sources/stacks-chow/tags.txt, source lines 4481–4481. Prerequisite pointer invoked at proof line(s) 650; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a simplicial topological space. Let $\mathcal{F}$ be an abelian sheaf on X. There is a spectral sequence $(E_r, d_r)_{r\geq 0}$ with*

$$E_1^{p,q} = H^q(X_p, \mathcal{F}_p)$$

converging to $H^{p+q}(X_{Zar}, \mathcal{F})$. This spectral sequence is functorial in $\mathcal{F}$.

Proof. Let $\mathcal{F} \to \mathcal{I}^\bullet$ be an injective resolution. Consider the double complex with terms

$$A^{p,q} = \mathcal{I}^q(X_p)$$

and first differential given by the alternating sum along the maps d_i^{p+1}-maps $\mathcal{I}_p^q \to \mathcal{I}_{p+1}^q$, see Lemma 09VM. Note that

$$A^{p,q} = \Gamma(X_p, \mathcal{I}_p^q) = \mathrm{Mor}_{PSh}(h_{X_p}, \mathcal{I}^q) = \mathrm{Mor}_{PAb}(\mathbf{Z}_{X_p}, \mathcal{I}^q)$$

Hence it follows from Lemma 09W5 and Cohomology on Sites, Lemma 03F6 that the rows of the double complex are exact in positive degrees and evaluate to $\Gamma(X_{Zar}, \mathcal{I}^q)$

in degree 0. On the other hand, since restriction is exact (Lemma 09W0) the map

$$\mathcal{F}_p \to \mathcal{I}_p^\bullet$$

is a resolution. The sheaves $\mathcal{I}_p^q$ are injective abelian sheaves on X_p (Lemma 09W1). Hence the cohomology of the columns computes the groups $H^q(X_p, \mathcal{F}_p)$. We conclude by applying Homology, Lemmas 0132 and 0133. □